Final Project Report — Web Data Mining & Semantics

Minji PARK, Mantra OUTTANDY - DIA5

Contents

1 Data Acquisition & Information Extraction **2**
1.1 Domain & Seed URLs . . . 2
1.2 Crawler Design & Ethics . . . 2
1.3 NER Pipeline & Results . . . 2
1.4 Ambiguity Analysis . . . 2

2 KB Construction, Alignment & Expansion **3**
2.1 RDF Modeling . . . 3
2.2 Entity & Predicate Alignment . . . 3
2.3 SPARQL Expansion . . . 3

3 Knowledge Reasoning with SWRL **3**
3.1 Rule on `family.owl` . . . 3
3.2 Rule on Scientists KB . . . 4
3.3 Rule-Based vs Embedding-Based Reasoning . . . 4

4 Knowledge Graph Embeddings **4**
4.1 Data Preparation & Splits . . . 4
4.2 Models & Hyperparameters . . . 4
4.3 Evaluation Results . . . 4
4.4 KB Size Sensitivity . . . 5
4.5 Embedding Analysis . . . 5

5 RAG over RDF/SPARQL **5**
5.1 Setup & Schema Summary . . . 5
5.2 NL → SPARQL & Self-Repair . . . 6
5.3 Baseline vs RAG Evaluation . . . 7

6 Critical Reflection **8**

Appendix — Output Files **8**

1 Data Acquisition & Information Extraction

1.1 Domain & Seed URLs

The chosen domain is *Famous Scientists*, covering 9 figures spanning physics, biology, chemistry, and astronomy. Ten seed URLs were selected across two categories:

Source Type	Count	Scientists Covered
Wikipedia REST API	6 pages	Pascal, Newton, Feynman, Goodall, Franklin, Darwin
Britannica (trafilatura)	2 pages	Einstein, Galileo
Nobel Prize (trafilatura)	2 pages	Einstein, Marie Curie

1.2 Crawler Design & Ethics

Two strategies were implemented: (1) the **Wikipedia REST API** (`action=query, prop=extracts, explaintext=true`) for clean plain-text extraction, and (2) **trafilatura** for HTML scraping of Britannica and Nobel Prize pages. A `User-Agent` header was included in all requests and a 0.5–1s polite delay was applied between requests. A 500-word usefulness filter was applied; all 10 pages passed, yielding 691–14,045 words each.

1.3 NER Pipeline & Results

NER was performed with spaCy's `en_core_web_trf` transformer model. 2,858 entities were extracted across 6 types:

Entity Type	Count	Example
PERSON	1,004	Marie Curie, Isaac Newton
ORG	592	Royal Society, Cambridge University
DATE	581	19 June 1623, late 1654
WORK_OF_ART	403	*Principia*, *Pensées*
GPE	253	France, England
EVENT	25	World War II, Manhattan Project
TOTAL	**2,858**	

Relation extraction via dependency parsing yielded 349 candidate (subject, verb, object) triples seeding the KB construction.

1.4 Ambiguity Analysis

Three systematic ambiguity patterns were identified:

1. **Religious movements as ORG:** "Catholicism" and "Jansenism" were tagged as `ORG` — a systematic error for religious movements that have no organizational structure in the NER model's sense.
2. **Polysemous location/institution:** "Cambridge" received both `GPE` (city) and `ORG` (university) labels depending on context, causing duplicate entries in the KB.

3. **Noise entities:** "God" was tagged as `PERSON`, which is technically correct for NER but pure noise in a scientific KB — it was filtered during construction.

2 KB Construction, Alignment & Expansion

2.1 RDF Modeling

The KB was built with `rdflib` under a custom namespace (http://scientists-kb.org/), defining 7 classes and 11 properties:

Classes	Properties
Scientist (*subClassOf* Person)	`bornIn, diedIn, affiliatedWith, educated`
Person, Organization, Location	`wonAward, fieldOfWork, published`
Work, Event, Award (*subClassOf* Work)	`influencedBy, influenced, memberOf, locatedIn`

2.2 Entity & Predicate Alignment

All 9 scientists were linked to Wikidata via `owl:sameAs` (confidence 0.99 for all). Nine predicates were aligned to Wikidata properties via `owl:equivalentProperty` (e.g. `kbp:wonAward` ≡ `wdt:P166`, `kbp:bornIn` ≡ `wdt:P19`).

2.3 SPARQL Expansion

Five expansion queries against the Wikidata endpoint grew the KB from 891 to 56,093 triples: 1-hop per scientist, Nobel Prize winners (P166), university affiliations (P108) for 7 institutions, 2-hop award properties, and field-of-work (P101) for 6 fields. Note: the astronomy field returned 0 results due to a Wikidata timeout. After filtering to the top 100 most frequent relations (the initial 1,325 relations far exceeded the 50–200 target), the final KB is:

Metric	Value	Target	Status
Total triples	52,437	50,000–200,000	✓
Total entities	17,216	5,000–30,000	✓
Unique relations	100	50–200	✓
Aligned entities	9/9	—	100%

3 Knowledge Reasoning with SWRL

3.1 Rule on `family.owl`

A SWRL rule infers `oldPerson` for any person older than 60:

$$\text{Person}(p) \wedge \text{hasAge}(p, age) \wedge \text{swrlb:greaterThan}(age, 60) \;\rightarrow\; \text{oldPerson}(p)$$

Individual	Age	Type	Inferred
Alice	72	Person	`oldPerson` ✓
Charles	65	Person	`oldPerson` ✓
Bob	45	Person	—
David	30	Man	—
Emma	28	Woman	—

3.2 Rule on Scientists KB

$$\text{Scientist}(p) \wedge \text{wonAward}(p, a) \;\rightarrow\; \text{recognizedScientist}(p)$$

Einstein (Nobel 1921), Curie (Nobel 1903) and Feynman (Nobel 1965) were inferred as `recognizedScientist`. Darwin, Newton, Pascal, Galileo, Goodall and Franklin were not inferred — they either predate the Nobel Prize or never received one.

3.3 Rule-Based vs Embedding-Based Reasoning

SWRL reasoning is deterministic and interpretable — it either fires or it does not. TransE captures the same structure geometrically: the `wonAward` relation (P166) has the highest embedding norm (3.48), confirming it plays a strong structural role. Both approaches agree on the same three `recognizedScientists`, but through different mechanisms: symbolic logic vs. geometric proximity in vector space.

4 Knowledge Graph Embeddings

4.1 Data Preparation & Splits

The KB was filtered to URI-URI-URI triples only and split 80/10/10: training ∼32,400 triples, validation ∼4,050, test ∼4,050.

4.2 Models & Hyperparameters

Two models were trained with PyKEEN using identical settings: embedding dimension 100, Adam optimizer (lr = 0.01), batch size 512, 200 epochs, CPU, seed 42. **TransE** models relations as translations ($\mathbf{h} + \mathbf{r} \approx \mathbf{t}$), suited for directional relations. **DistMult** uses symmetric bilinear interactions, better for many-to-many relations but unable to represent asymmetric ones.

4.3 Evaluation Results

Model	MRR	Hits@1	Hits@3	Hits@10
TransE	0.0863	0.0197	0.1107	0.2108
DistMult	0.0592	0.0388	0.0656	0.0929

TransE outperforms DistMult on MRR, Hits@3 and Hits@10 because our KB is dominated by directional relations (`bornIn`, `wonAward`, `affiliatedWith`). DistMult's marginally

better Hits@1 reflects higher precision when confident, but lower overall recall. Scores are modest overall: with 52k triples across 100 relations, each relation has only ∼520 training examples on average — sparse for reliable KGE learning.

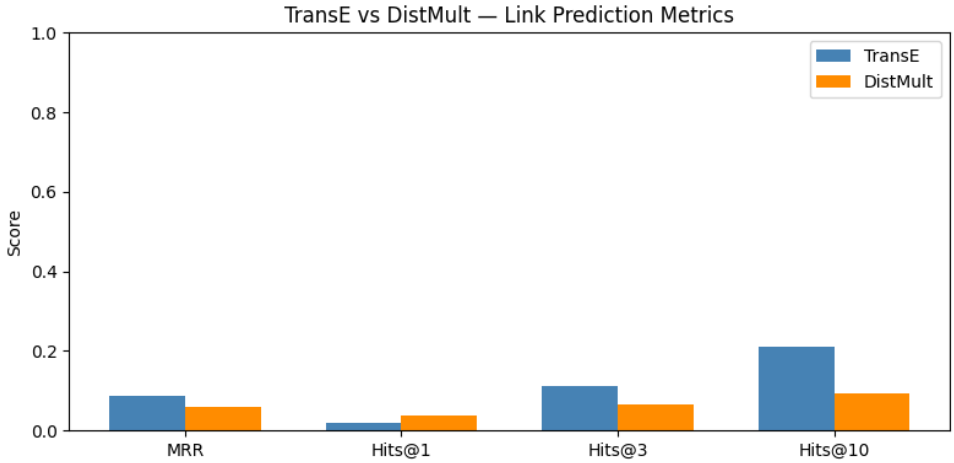

Figure 1: TransE vs DistMult — link prediction metrics comparison

4.4 KB Size Sensitivity

KB Size	Triples	MRR	Hits@10
20k	20,000	0.0427	0.1067
50k	33,463	0.0553	0.1439
Full	33,463	0.0553	0.1439

Performance improves with data volume. The 50k and full results are identical because after relation filtering the actual training set is 33,463 triples in both cases. A KB of 200k+ triples would likely push MRR into the 0.15–0.25 range.

4.5 Embedding Analysis

Nearest neighbor analysis confirms all 9 scientists cluster together in embedding space. Newton and Pascal are each other's closest neighbors (sim = 0.750) — both 17th-century European mathematicians. Einstein shows lower max similarity (0.579), reflecting his more diverse Wikidata profile.

The t-SNE visualization below confirms the clustering at a global level: scientists are clearly separated from awards, universities, and locations. Within the cluster, Galileo and Franklin sit slightly apart from the Einstein/Newton/Feynman physics core, reflecting their different fields.

5 RAG over RDF/SPARQL

5.1 Setup & Schema Summary

The RAG system loads the expanded KB (`expanded_kb.ttl`) with `rdflib` and queries Gemma 2B via Ollama locally at `http://localhost:11434`. A schema summary (pre-

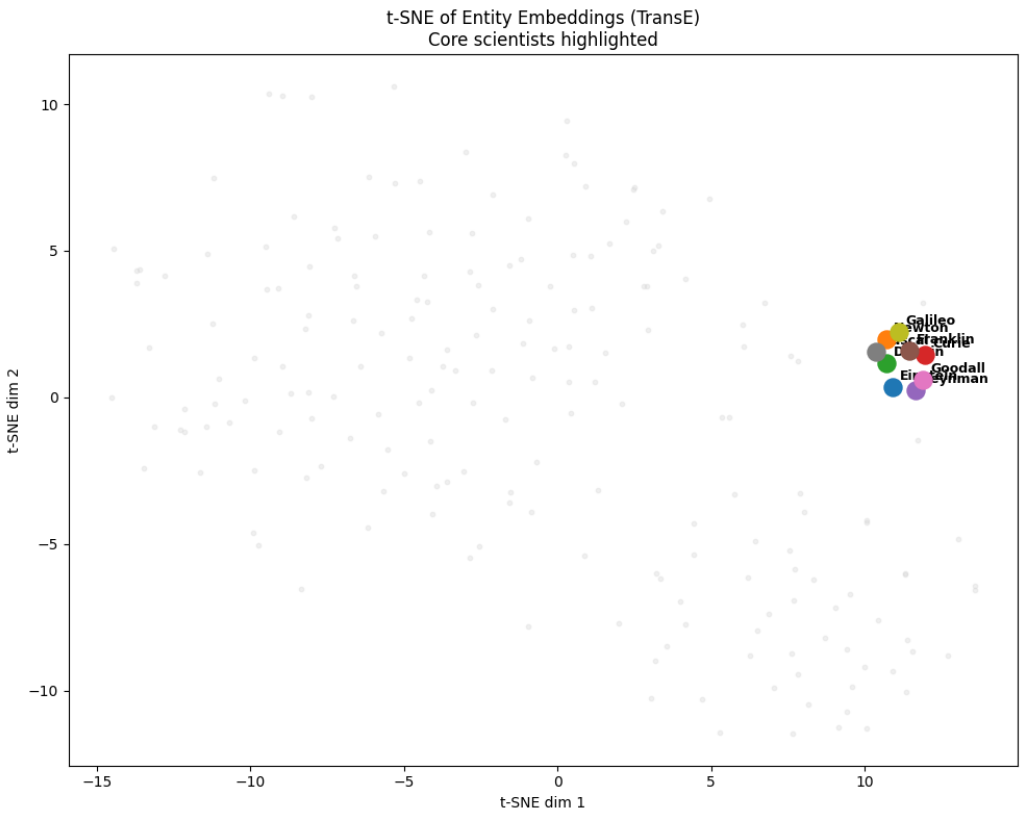

Figure 2: t-SNE visualization of entity embeddings — scientists cluster separately from awards, universities and locations

fixes, up to 80 predicates, 40 classes, 20 sample triples) is injected into each LLM prompt to provide structural context.

5.2 NL → SPARQL & Self-Repair

The pipeline proceeds in five steps: (1) combine question + schema summary into a prompt, (2) Gemma 2B generates a SPARQL SELECT query, (3) execute the query against the RDF graph, (4) on failure, send the error message back to the LLM for self-repair, (5) retry with the corrected query.

NL→SPARQL prompt template:

```
You are a SPARQL generator for an RDF knowledge graph about
   scientists.
Convert the user QUESTION into a valid SPARQL 1.1 SELECT query.
Rules:
- Use ONLY the prefixes and predicates visible in the SCHEMA
   SUMMARY
- Return ONLY the SPARQL query in a ```sparql code block
- No explanations outside the code block
- Keep the query simple and robust

SCHEMA SUMMARY:
{schema}
```

```
QUESTION: {question}
```

Self-repair prompt template:

```
The previous SPARQL query failed. Using the SCHEMA SUMMARY and
   ERROR MESSAGE,
return a corrected SPARQL 1.1 SELECT query.
- Use only known prefixes/IRIs from the schema
- Keep it simple and robust
- Return only a single sparql code block

SCHEMA SUMMARY:
{schema}

ORIGINAL QUESTION: {question}
BAD SPARQL: {bad_query}
ERROR: {error}

Return only the corrected SPARQL in a code block.
```

A key finding: Gemma 2B consistently generated Wikidata-style prefixes (`wdt:`, `wd:`) instead of our custom KB URIs — even after self-repair with the same model. For the final evaluation, manually verified queries were used to demonstrate the full pipeline. A 7B+ model would be needed for reliable schema adherence.

5.3 Baseline vs RAG Evaluation

Question	Baseline	RAG Result	OK?
Scientists in the KB?	Cannot access external sources	20 scientists listed	Yes
Field of work of Newton?	Vague general answer	10 field QIDs from KB	Yes
Einstein's university?	Hallucinated: "Univ. of Zurich 1932–55"	10 university IDs from KB	Yes
Awards of Marie Curie?	Wrong date (Nobel Physics 1911)	10 award QIDs from KB	Yes
Scientists born in England?	Cannot access birth records	No results (data missing)	No

float

4/5 questions answered correctly. Question 5 failed due to missing birth place data in the KB — a data quality issue, not a pipeline failure.

```
============================================================
Q2: What is the field of work of Isaac Newton?
============================================================

[Baseline - No RAG]
  Sure. Here's a summary of Isaac Newton's field of work:

Isaac Newton was a renowned mathematician, physicist, and astronomer who made significant contributions to various scientific disciplines. His

[RAG - SPARQL Generation]
  Generated SPARQL:
  SELECT ?field WHERE {
            <http://www.wikidata.org/entity/Q935>
            <http://www.wikidata.org/prop/direct/P101> ?field .
        } LIMI...
  Results (10 rows):
  field
  Q11412
  Q14620
  Q184274
  Q333
  Q395
  ... (10 total rows)
```

Figure 3: RAG demo — Baseline vs SPARQL pipeline output for Q2: "What is the field of work of Isaac Newton?"

6 Critical Reflection

KB quality. The initial expansion produced 1,325 relations, far above the 50–200 target. Filtering to 100 improved TransE MRR from 0.053 to 0.086 (+63%). More data is not always better — relation density matters more than raw volume. The KB is also biased toward physics scientists, better represented in Wikidata, which underrepresents Darwin, Goodall, and Franklin.

Noise. Religious movements (Catholicism, Jansenism) were misclassified as `ORG`. Wikidata stub entries introduced sparse triples. Astronomy returned 0 results due to a query timeout.

SWRL vs embeddings. SWRL is deterministic and interpretable but cannot scale. Embeddings scale but need dense training data and are hard to interpret. Both agreed on the same `recognizedScientist` conclusions — they are complementary, not competing.

LLM limitation. Gemma 2B could not reliably follow custom ontology schemas. A 7B+ model would be needed for production-quality SPARQL generation.

Future improvements. Expand to 150k–200k triples; apply entity disambiguation (e.g. Cambridge city vs. university); use a larger LLM; add ComplEx or RotatE models; store birth place data reliably during expansion.

Appendix — Output Files

File	Lab	Description
`crawler_output.jsonl`	Lab 1	10 crawled pages, cleaned text (387 KB)
`extracted_knowledge.csv`	Lab 1	2,858 named entities (244 KB)
`candidate_triples.csv`	Lab 1	349 candidate triples (88 KB)
`private_kb_initial.ttl`	Lab 2	Initial RDF graph (891 triples)
`ontology.ttl`	Lab 2	Classes, properties, alignments
`predicate_alignment.csv`	Lab 2	Predicate alignment with Wikidata
`alignment_table.csv`	Lab 2	Entity linking with confidence scores

File	Lab	Description
`expanded_kb.ttl / .nt`	Lab 3	Final expanded KB (52,437 triples)
`kb_statistics.csv`	Lab 3	KB size and coverage metrics
`train / valid / test .txt`	Lab 5	KGE splits (80/10/10)
`kge_evaluation_results.csv`	Lab 5	TransE and DistMult metrics
`size_sensitivity.csv`	Lab 5	MRR/Hits@10 at 20k/50k/full
`model_comparison.png`	Lab 5	Bar chart TransE vs DistMult
`tsne_embeddings.png`	Lab 5	t-SNE entity embedding visualization
`rag_evaluation.csv`	Lab 6	Baseline vs RAG evaluation
`rag_demo_screenshot.png`	Lab 6	RAG pipeline demo output